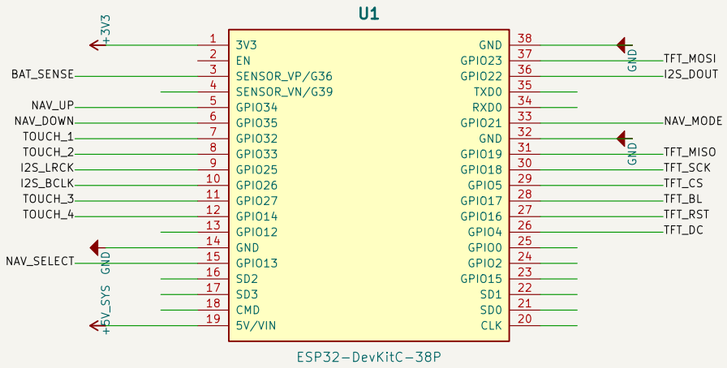
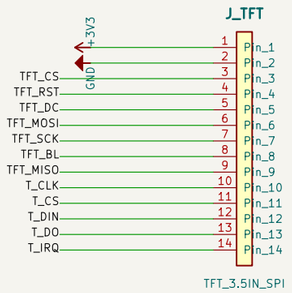
BLOCK 1: ESP32-WROOM-32 MICROCONTROLLER CORE

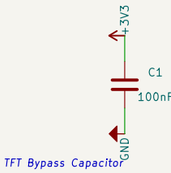
Standard 38-Pin DevKitC Module socketed via 2x 1x19 2.54mm female headers



BLOCK 2: 3.5-INCH SPI COLOR TFT DISPLAY (480x320)

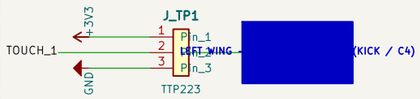
14-Pin Header Socket directly above ESP32; Hardware SPI Bus + Backlight PWM

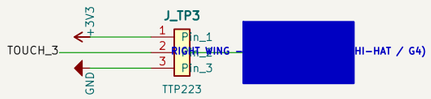


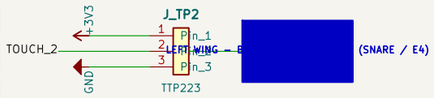


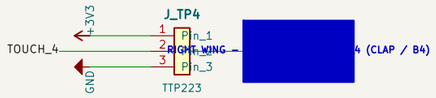
BLOCK 3: SPLIT 2-LEFT / 2-RIGHT CAPACITIVE TOUCH CONTROLS

Four TTP223 Modules plugged into 1x3 sockets + Large Copper Touch Plates



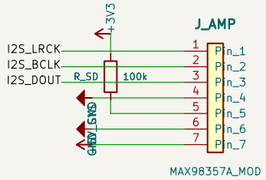


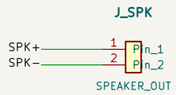




BLOCK 4: I2S AUDIO SUBSYSTEM

MAX98357A Mono Class-D Amplifier + Speaker Terminal

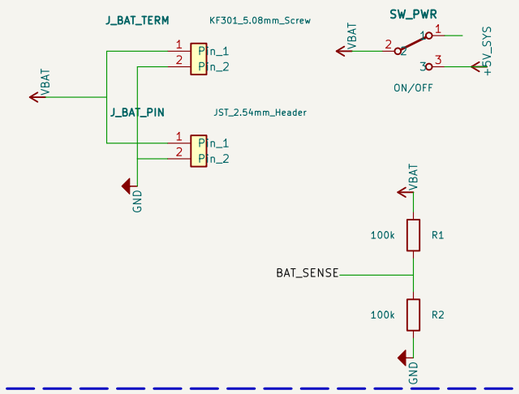




Connect to 4 Ohm / 8 Ohm 3W Micro-Speaker

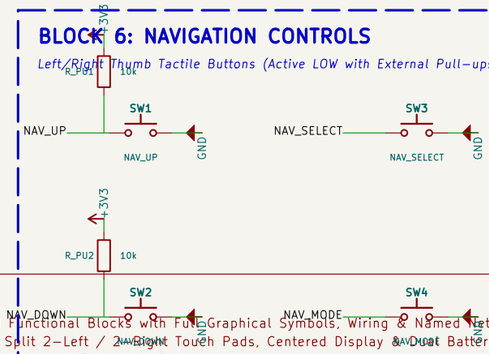
BLOCK 5: UNIVERSAL DUAL BATTERY INPUT & POWER

5.08mm Screw Block + 2.54mm Header in Parallel; Slide Switch & ADC Divider



BLOCK 6: NAVIGATION CONTROLS

Left/Right Thumb Tactile Buttons (Active LOW with External Pull-ups)



Organized Functional Blocks with Full Graphical Symbols, Wiring & Named Net Labels
Option 1: Split 2-Left / 2-Right Touch Pads, Centered Display & Dual Battery Ports

Custom Hardware & Embedded Systems

Sheet: /
File: music_pad.kicad_sch

Title: ESP32 Handheld Cyber-Console Music Pad

Size: A3	Date: 2026-10-19	**Rev: v2.0 - Handheld Gaming**
KiCad E.D.A. 10.0.6		Id: 1/1